Computer Programming-II Laboratory (IMCA – 2nd semester)

Experiment List

Experiment-01

1. Write a program to take a string as input and print it.
2. Create a program to input a string and display each character on a new line.
3. Write a program to reverse a given string and display it.
4. How can you implement a program to count the number of vowels in a string?
5. Create a program to input a sentence and display the total number of words.
6. Write a program to replace spaces in a string with hyphens.
7. Develop a program to count occurrences of a specific character in a string.
8. How can you write a program to find the longest word in a sentence?
9. Develop a program to merge two strings alternately (e.g., "abc" and "123" → "a1b2c3").

Experiment-02

1. Develop a program to concatenate two strings using strcat.
2. Write a program to compare two strings using strcmp.
3. Create a program to find the length of a string using strlen.
4. Write a program to copy one string to another using strcpy.
5. Write a program to check if a string is a palindrome using strrev.
6. Develop a program to convert a string to uppercase using strupr.
7. Write a program to convert a string to lowercase using strlwr.
8. Implement a program to check if a substring exists within a string using strstr.
9. Write a program to demonstrate the use of substr()function.
10. Write a program to demonstrate the use of atoi()function.

Experiment-03

1. Write a program to define a structure for a student containing fields for name, roll number, and marks. Input and display the details.
2. How can you define a structure to store a date (day, month, year) and display it in the format DD/MM/YYYY?
3. Write a program to define a structure for a complex number (real and imaginary parts) and write a function to perform addition of two complex numbers.
4. Create a nested structure to store a student's details (name, roll number, and address where address is another structure).
5. Demonstrate the use of bit fields with structure to store specific information compactly (e.g., flags or settings).
6. Demonstrate: How to use typedef with structures to simplify code readability?

7. Write a program to store details of multiple employees and count the number of employees in a specific department using nested structures.
8. Develop a program to create a structure for a 2D point and use it to calculate the distance between two points.
9. Write a program to define a structure for a car (make, model, and year) and use nested structures to include the owner's details.

Experiment-04

1. Demonstrate: How do you declare a bit-field to store a value that ranges from 0 to 15?
2. Write a program to use enumeration to represent days of the week.
3. Write a C program that uses an enumeration to implement a menu-driven application.
4. Can a union contain a bit-field? Demonstrate with an example.
5. How can you initialize an array of structures? Show an example.
6. Write a program to define a structure for a book with title, author, and price. Input details for 10 books and display them.
7. Write a program to define a structure within another structure to store details of an employee (name, ID, salary and department details). Store the details of 10 employees and sort them according to the salary.
8. Write a program to define a structure for a student containing fields for name, roll number, and marks. Input the details of 10 students and display the details of the student who has secured highest mark.

Experiment-05

1. Write a program to create a pointer to an integer variable and demonstrate how the pointer can modify the value of the variable.
2. Explain how you can use a pointer to access and print elements of an integer array.
3. Create a program to illustrate the use of pointers with character variables. Include both reading and printing the character via the pointer.
4. Write a program demonstrating pointer arithmetic with a double array. Show how to iterate through the array using a pointer.
5. How would you use a pointer to change the value of a variable of type `long`? Create a program to showcase this concept.
6. Write a program that passes an integer variable by reference to a function using a pointer. Modify the value in the function and print it in the main program.
7. Create a program where a pointer to a pointer (e.g., `**p`) is used to access and modify a variable of type `short`.

Experiment-06

1. Write a program to input an array of integers and print them using pointers.
2. Implement a function to find the largest element in an array using pointers.
3. Write a program to reverse the elements of an array in-place using pointers.
4. Implement a function to calculate the sum of all elements in an array using pointers.
5. Create a function to sort an array in ascending order using pointers.
6. Write a program to find the frequency of a given element in an array using pointers.
7. Write a program to rotate the elements of an array to the left by one position using pointers.
8. Write a program to count the number of even and odd elements in an array using pointers.

Experiment-07

1. Write a program to calculate the length of a string using pointers.
2. Write a program to copy one string to another using pointer.
3. Implement a function to compare two strings using pointers.
4. Implement a function to count the number of vowels in a string using pointers.
5. Implement a function to convert all lowercase characters in a string to uppercase using pointers.
6. Write a program to find the frequency of a given character in a string using pointers.
7. Implement a function to count the number of words in a string using pointers.
8. Write a program to shift each character in a string by a given number (Caesar Cipher) using pointers.

Experiment-08

1. Write a program to create a structure named Complex and define a function incrComplex() to increment the value of both real and imaginary members by 1.
2. Define a structure Distance to store distance in kilometer, meter & centimeter. Create an array of Distance type to store 5 Distance variables. Find the longest distance using pointers.
3. Define a function swapDistance() for the structure defined in question no.2, to swap the content of two Distance variables.
4. Define 2 functions producer() and consumer() which will operate on a common variable item. Producer () will increment the variable item & consumer () will decrement the value of item. Producer cannot increment if item reaches at 10. Consumer () cannot decrement when item reaches 0.

5. Create structure to store details of book. Store and print the details of 5 books using pointer.

Experiment-09

1. Write a program to swap two numbers using call by reference.
2. Write a program to reverse a string using call by reference.
3. Create a program to convert all characters of a string to uppercase using call by reference.
4. Write a function to sort an array of integers in ascending order using call by reference.
5. Create a program to calculate the transpose of a matrix using call by reference.
6. Implement a program to reverse the digits of a given integer using call by reference.
7. Create a program to remove all spaces from a string using call by reference.

Experiment-10

1. Implement a function to calculate the average of elements in a 1D array.
2. Implement a function to search for a specific element in a 1D array.
3. Create a function to add two 2D matrices.
4. Write a program to multiply two 2D matrices using a function.
5. Write a program to compare two structures using a function.
6. Create a function to calculate the sum of salaries of employees stored in an array of structures.
7. Write a program to copy the contents of one structure to another using a function.

Experiment-11

1. Write a function to calculate the length of a string without using the strlen function.
2. Create a function to compare two strings for equality.
3. Write a function to convert all characters in a string to lowercase.
4. Design a function to count the number of vowels in a string.
5. Implement a function to replace all spaces in a string with underscores.
6. Write a function to check if a string is a palindrome.
7. Create a function to remove all instances of a character from a string.
8. Write a function to find the longest word in a string.

Experiment-12

1. Write a program to print all command-line arguments passed to the program.
2. Create a program to calculate the sum of integers passed as command-line arguments.
3. Design a program that checks if the number of command-line arguments meets specific criteria (e.g., minimum number of arguments).
4. Create a program to count the number of command-line arguments passed.
5. Write a program that accepts a string as a command-line argument and converts it to uppercase.
6. Write a program using #define to declare constants and use them in arithmetic operations.
7. Create a program that demonstrates the use of #include to import custom header files.
8. Write a macro to calculate the square of a number.
9. Implement a macro to determine the maximum of two numbers.

Experiment-13

1. Implement a program to read content from a text file and print it to the console.
2. Write a program to count the number of lines, words, and characters in a given text file.
3. Implement a program to append text to an existing text file.
4. Write a program to compare two text files and check if their contents are identical.
5. Create a program to copy the content of one text file into another.

Experiment-14

1. Write a program to create a binary file and store the details of 5 students (name, roll number, and marks) and display them on the screen.
2. Write a program to append data to an existing binary file without overwriting the existing data.
3. Write a program to search for a particular record (e.g., a student's details) in a binary file and display it.